Wattmeter



Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Power Meter; Watt meter, pressure meters; Wattmeter; Wattmeter and Clamp meter

What it is used for

A wattmeter measures electrical power consumed or delivered by a circuit or device. Correct voltage and current connections, suitable ratings, and careful interpretation help prevent shock and inaccurate measurements.

Main parts

- Display or scale shows measured power and related values.
- Voltage input terminals connect across the measured supply or circuit.
- Current input terminals connect in series or through the specified current sensor.
- Current coil or sensor measures current in applicable meter designs.
- Selector controls choose measurement mode or range.
- Test leads, probes, or clamp sensor connect the instrument to the circuit.
- Protective casing and rated insulation help reduce electrical hazards.

Before you start

1. Read the instrument manual and confirm its voltage, current, and power ratings suit the circuit.
2. Inspect the casing, test leads, probes, connectors, and clamp sensor for damage.
3. Identify the supply type and the instrument's required voltage and current connection method.
4. Switch off and isolate the circuit before making connections whenever the procedure permits.
5. Use the correct terminals and range according to the wiring diagram.
6. Wear suitable PPE and obtain authorised supervision for mains or high-energy circuits.

How to use

1. Set the instrument to the required power measurement mode and range.
2. Connect the voltage inputs across the specified circuit points according to the manual.
3. Connect current inputs in series or position the clamp sensor as specified for the model.
4. Check polarity, phase, and connection direction where required by the instrument.
5. Verify all connections are secure before energising the circuit for measurement.
6. Read the wattage and other displayed values after the reading stabilises.
7. Check that the reading is within the instrument's rated range and expected operating conditions.
8. Record the value with units and relevant AC or DC details.
9. Switch off and isolate the circuit before disconnecting leads whenever possible.
10. Remove the connections carefully and store the instrument and accessories safely.

⚠ Safety

- Never exceed the instrument's voltage, current, or measurement-category ratings.
- Do not connect current terminals directly across a voltage source unless the manual explicitly permits it.
- Never handle exposed conductors or damaged leads on an energised circuit.
- Keep fingers behind probe guards and use suitable protective equipment.
- Use clamp sensors only according to their specified conductor and current limits.
- Do not change wiring or terminals while the circuit is live.

Care & storage

- Switch off and disconnect the instrument before cleaning or inspecting accessories.
- Clean the casing with a dry, suitable cloth and keep terminals free from dirt.
- Inspect leads, connectors, clamp jaws, and insulation regularly for wear or damage.
- Follow the manufacturer's instructions for battery replacement, calibration, and servicing.

Common mistakes to avoid

- Connecting voltage and current inputs to the wrong terminals.
- Exceeding the instrument's rated current or voltage range.
- Using a clamp sensor incorrectly or around multiple conductors when a single conductor is required.
- Recording a reading without checking units, mode, or connection direction.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar